

Characterizing space-use of a free-flying colony of Egyptian fruit bats (*Rousettus aegyptiacus*)
for ex-situ captive management

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Project Duration: February – December 2026

Project Category: Research; Population Management

Abstract

Understanding how managed zoo populations of animals utilize habitat can be referred to as space use. Space use, a fundamental component of characterizing animal welfare within captive conditions, can be used to inform long-term exhibit design and ethical conservation programs that depend on ex-situ conservation. While numerous studies acknowledge these nuances in large, charismatic species (e.g., non-human primates) where behavioral patterns are easily observed, effective management of all zoo animal populations would benefit from more rigorous research. By leveraging the largest, free-flying population of captive bats at Henry Doorly Omaha Zoo, we may construct a comprehensive inquiry into how human visitation influences use of a large habitat. This study utilizes a camera-trapping monitoring survey to quantify exhibit space use by free-roaming bats to provide a framework for welfare assessment in a lesser-studied captive taxa.

Project Overview

Space use, how animals distribute themselves within their environment, is a powerful observational tool to characterize how animals behave in captive, managed populations. Historically, research has investigated how space use may reveal insights into zoo habitat quality, which has direct influences on animal behavioral health (Ross *et al.* 2009). Over the last decade, numerous studies, particularly focusing on non-human primates, have identified meaningful impacts poor exhibit design and anthropogenic disturbance (coined the “visitor effect”) has on long-term space use (Earl, Hopper and Ross 2020; Duncan, D’Egidio Kotze and Pillay 2022; Wark *et al.* 2023). Collectively, these studies acknowledge the intense need to quantify space-use patterns to gauge captive animal wellbeing.

However, most space-use and visitor-effect studies focus on large, charismatic species. Many small, mammalian taxa are poorly understood – particularly bats. Existing work on captive fruit bats have detailed flight patterns (Bell *et al.* 2019), social activity (Welch *et al.* 2020) and enrichment priorities (LeBlanc, unpublished). These studies, however, have limited input on how bats utilize exhibits at a fine scale, or how they react to human visitation. This project aims to address this gap by providing camera trap monitoring to quantify fine-scale space use by free-roaming fruit bats, housed in a mixed-use exhibit. By recording motion-activated behavioral data, we may investigate how a breadth of environmental habitat factors can influence space-use in a volant captive mammal. Because human presence is shown to alter behavior of zoo animals

(Sherwin, Montgomery and Lundy 2013), understanding these dynamics across all taxa is critical to a One-Plan approach to ethical ex-situ management.

Overarching Goal

The overarching goal of this project is to quantify how free-ranging fruit bats use exhibit space in relation to human presence and other animal activity within a mixed-use zoo environment. We aim to leverage these behavioral data to promote evidence-based ex-situ management.

Methods

To take stock of how captive *Rousettus aegyptiacus* utilize a large (approximately 49,500 square foot) free-flying enclosure (Lied Jungle, Henry Doorly Omaha Zoo, Omaha, NE, USA), an assemblage of PR801W 8K Solar Wi-Fi trail cameras (Shenzhen Shangongda Trade Co., LTD, CN) will be mounted to non-enclosed, human-accessible structures. Trail cameras will be mounted at varying heights to account for high-and-low bat flight and other animal activity. Cameras will be configured to record 10 second clips upon PIR (motion-detected) activation. We will BLOCK two separate recording timeframes, with half of the total surveys occurring on weekdays (when human visitation is expected to be moderate), with the other half on weekends (when human visitation is expected to be high), with surveys occurring in the Spring.

All video clips will be manually reviewed by technicians. Technicians will record observational variables, including, a) bat presence (yes/no), b) human presence (yes/no), and c) presence of other free-roaming animals (yes/no). For each clip, attached data will include time of recording (TOR), date, average hourly temperature (recorded by zoo staff) and camera ID. Camera ID will be attributed to each sampling “cell” (Figure 1C). Summary data (# of bat clips, # of human clips, # other clips) will be totaled for each sampling cell (Figure 1C).

For the purposes of quantifying bat use of each cell, we will treat the count of bat clips (where bats are confirmed active) as the response variable. Tailored for count data, we will model bat activity using generalized linear models (GLMMs) with a negative binomial error distribution, with a log offset term for survey effort ($\log(\text{total clips hourly clips})$). Models will be compared with the Akaike Information Criterion for small datasets (AICc). The model with the best fit will be used to determine the influences driving free-flying bat space-use.

Timeline

February, March 2026: Purchase equipment, higher technicians

Early April 2026: Test-run camera deployment placement, recording, etc.

May – August: Run 8 weekday, 8 weekend sampling; biweekly

September: Analyze video, enter data

October: Model selection

November – December: dissemination of results

Dissemination of Results

Results will be presented via a publication, aimed to address extending behavioral research to understudied and zoo-managed taxa. All technicians, zoo collaborators, and volunteers will be invited as coauthors.

Budget

Item	Description	Quantity	Unit Cost (USD)	Total (USD)
PR801W Trail Cameras	Core data-collection devices for spatial monitoring	10	\$300	\$3,000
SD Cards (128–256 GB)	Long-term high-resolution video storage	10	\$25	\$250
Heavy-Duty Camera Mounting Hardware	Wall/beam mounts compatible with indoor enclosures	10	\$40	\$400
Visitor Signage	Informational signs noting active video monitoring	10	\$30	\$300
Technician Salary	1 technician for 100 hours @ \$25/hr (or split between two techs)	100 hrs	\$25/hr	\$2,500
Data Storage / Backup (Portable SSDs)	Backup for multi-week video load	2	\$150	\$300
Travel to/from Omaha's Henry Doorly Zoo	Regional travel (Nebraska–Iowa area), fuel + mileage reimbursement for recurring visits	—	—	\$1,200
Software Licenses / Cloud Backup (Optional)	If using paid cloud storage or analysis tools	1	\$300	\$300
Contingency (10–15%)	Replacement parts, batteries, unexpected repairs	—	—	\$2,500

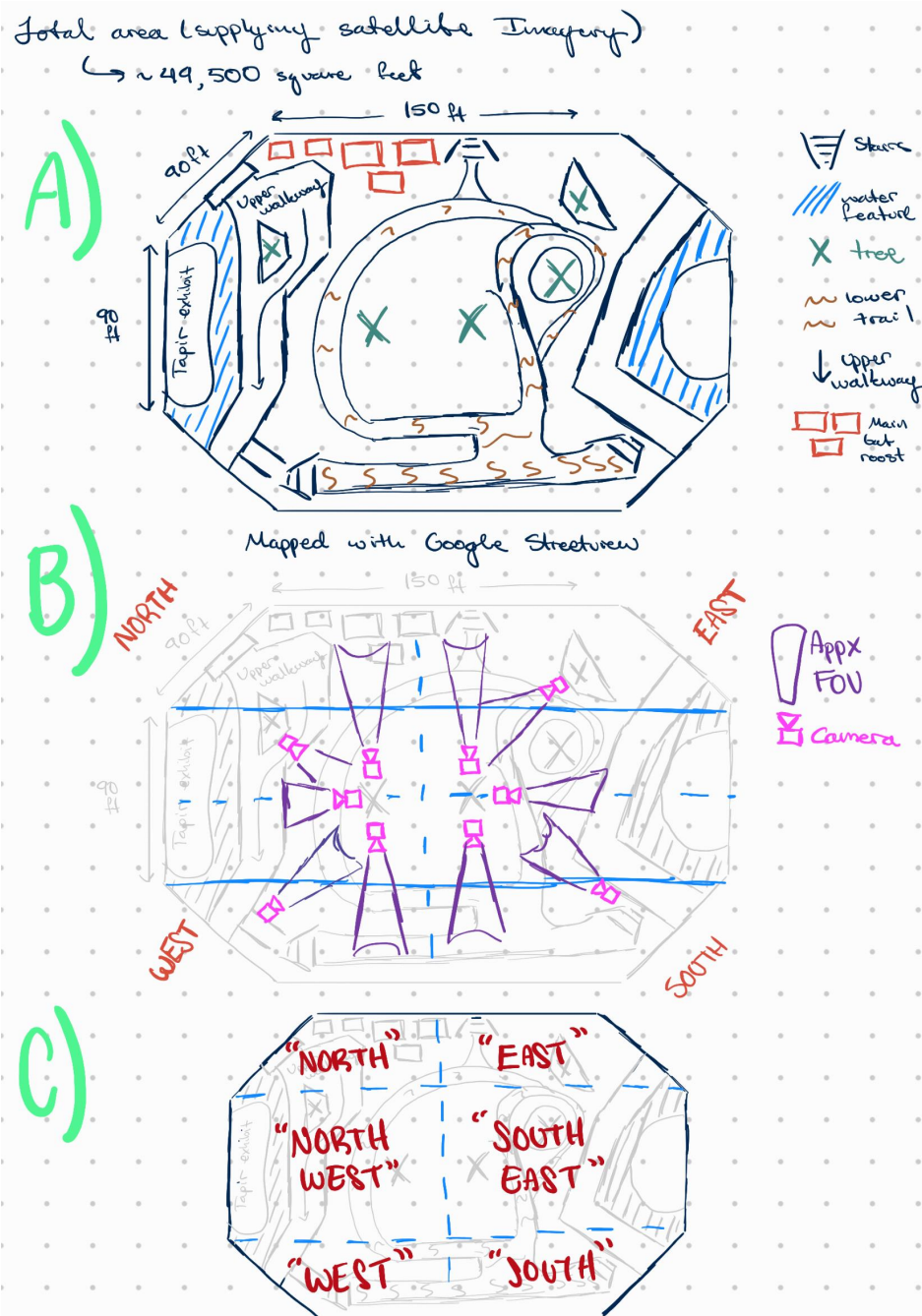


Figure 1 (sections A, B, and C). Section A provides a general schematic of the Lied Jungle exhibit, detailing walkways, water features, mountable surfaces (e.g., trees), and identifies where bats have been historically observed in large numbers (main bat roost, as above). Section B details the placement and relative FOV of each camera. Section C denotes the names of each survey cell.

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